Probability By Inquiry

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2026-06-09

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Welcome to PbI

This is the website for *Probability By Inquiry* by Jessie Oehrlein.

Source files for this book can be found on GitHub at github.com/jessoehrlein/Probability-By-Inquiry.

What Is This Book?

This book is intended for a calculus-based undergraduate probability theory course that uses inquiry-based learning.

What Is Probability Theory?

Probability theory is the area of mathematics that defines and proves ideas about probability.

If you think it's cheating to use "probability" in that definition… well, yes, it kind of is. So what is probability, anyway? Depending on the context and your philosophy, you might think of probability in a variety of ways. Here are two common interpretations:

- How often an outcome will occur if we measure a process over many repetitions or a long period of time;
- How likely we believe a certain outcome is.

Our mathematical tools will allow us to say some things about probability that are true regardless of these interpretations. Instead, we'll start from thinking of probability as a function that follows certain rules, and then we'll figure out what happens from there.

That doesn't mean that real world scenarios and interpretations will have no effect on the math we do! The particular "shapes" of probability we'll look at are shaped by their applicability to common scenarios. None of this math sprang out of nowhere. Most of it was developed because people had uses for it.

What Is Inquiry-Based Learning?

Inquiry-Based Learning in mathematics is a teaching & learning approach in which you, the student, do most of the mathematical sense making in collaboration with your peers and instructor. The instructor serves as a guide through this mathematical landscape, and their goal is for you to discover or explore certain ideas, but student thinking is the central focus of the course. CITATION HERE.

For this book, that means that **you**, along with your peers, are going to build up the key ideas and results of probability theory. Each chapter is structured as a sequence of definitions,

exercises, and theorems. There are very occasional examples, and there are not solutions. You will:

- Carefully read the **definitions**. These often also introduce new notation. When you encounter a new definition, it's a good idea to write down an example for yourself. In some cases, a non-example might also be useful! Write down any questions you have about the definition, too.
- Solve the **exercises**. These apply the definitions that have been introduced. As you go through the book, they'll also build more and more on the previous ideas you've figured out!
- Prove the **theorems**. Theorems are statements about something that is true in general. You will write proofs that build on definitions, axioms, and prior results to explain why these statements must be true. Think of the audience of these proofs as slightly skeptical math students at a similar level. What level of detail would help them to be convinced?

At the end of each chapter, you'll find sections titled Further Practice and Challenge Questions. Further Practice sections include exercises that are meant to help you be more comfortable with the definitions and tools of the chapter. Like the name indicates, they're more practice at a similar level. Challenge Questions sections include exercises or proofs that push the ideas a bit further.

What Should I Already Know?

This book assumes that you are familiar with single-variable calculus and are comfortable with functions, derivatives, integrals.

It does not assume that you have previously seen set theory, counting, or statistics or that you have previous experience writing proofs, though those will make various sections of the book a bit easier!

Some multivariate calculus is helpful in CHAPTER NAME HERE, but APPENDIX NAME HERE has a crash course with everything you need to know.

1 Set Theory

1.1 Introduction

We'll define probability as a function that follows some rules. But functions need some kind of input and output, so we need to start there. We'll say that a function maps from a **set** X to a set Y. This chapter is all about what we mean by a **set** and how these sets behave.

1.2 Definitions, Exercises, and Theorems

> **Note 1: Set**
>
> A **set** A is an unordered list of elements. We commonly describe A's elements in two ways.
>
> - **Roster notation** lists the elements: $\{a_1, a_2, \cdots a_n\}$
> - **Set-builder notation** uses a condition that exactly describes the elements of A. For example, the set of positive integers less than 20 could be written as $A = \{a |\ a \in \mathbb{Z}^+, a < 20\}$.

> **Note 2: Element and Subset**
>
> If t is an element of set A, then we say that t is an **element** of A, that t is **in** A, or $t \in A$. If every element of a set B is also in A, then we say that B is a **subset** of A, or $B \subseteq A$.

For example, $\{2, 4\}$ is a subset of $\{1, 2, 3, 4, 5\}$, but $\{4, 6\}$ is not. The empty set $\emptyset$ (the set of no elements) is considered a subset of all sets. A set is also considered a subset of itself.

Exercise 1.1. Let $A = \{n \mid n = 4k+1, k \in \mathbb{Z}\}$ and let $B = \{2, 5, 8, 11, 14, 17, 20\}$. Choose and write a non-empty subset of A in roster notation, and choose and write a non-empty subset of B in set-builder notation.

Exercise 1.2. Show that $A \subseteq C$ where $A = \{n \mid n = 4k + 1, k \in \mathbb{Z}\}$ and $C = \{m \mid m = 4l - 3, l \in \mathbb{Z}\}$.

Note 3: Set Equality

We say that two sets A and B are **equal**, or $A = B$, if they have exactly the same elements. This is equivalent to both $A \subseteq B$ and $B \subseteq A$ being true.

Exercise 1.3. Use the definition of set equality to show that $A = C$ where $A = \{n \mid n = 4k + 1, k \in \mathbb{Z}\}$ and $C = \{m \mid m = 4l - 3, l \in \mathbb{Z}\}$. (This should partially use work you've already done. There are other ways to show this, but this exercise is about practicing using the definition!)

Note 4: Union and Intersection

Given sets A and B, their **union** $A \cup B$ is the set of all elements that are in at least one of A or B. Given n sets $A_1, A_2, \cdots, A_n$, their union $\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup \cdots \cup A_n$ is the set of all elements that are in at least one of the original n sets.
The **intersection** of A and B, denoted $A \cap B$, is the set of all elements that are in both A and B. The intersection of n sets $\bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap \cdots \cap A_n$ is the set of all elements that are in every one of the original n sets.

For example, $\{1, 2, 3, 4, 5\} \cup \{4, 6\} = \{1, 2, 3, 4, 5, 6\}$, and $\{1, 2, 3, 4, 5\} \cap \{4, 6\} = \{4\}$.

Exercise 1.4. Find the intersection of A and B from Exercise 1.1. Find the union of the subsets of A and B that you described in Exercise 1.1.

Exercise 1.5. Define three sets A_1, A_2, and A_3 of your choosing, and find their union $\bigcup_{i=1}^{3} A_i$ and their intersection $\bigcap_{i=1}^{3} A_i$.

Exercise 1.6. Show that $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$.

Theorem 1.1 (Distributive Law)**.** *Prove the following visually, and then prove them using the definition of set equality (Definition 3). (You're already halfway there on the first statement with Exercise 1.6!)*

For any sets A, B, and C, we have that

- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$;
- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Note 5

We call a set S the **universal set** if it is the set of all elements that we are considering to be possible in a particular context.
Given a set $A \subseteq S$, the **complement** of A, denoted A^C or $\bar{A}$, is the set of all elements of S not in A.
Given two sets A and B, the **difference** $A - B$ is the set of all elements of A that are not in B.

Exercise 1.7. Let $S = \{0, 2, 4, 6, 8, 10\}$, $A = \{0, 4, 8\}$, and $B = \{0, 2, 8, 10\}$. Find A^C, B^C, $A - B$, and $B - A$.

Theorem 1.2 (DeMorgan's Laws)**.** *Prove the following for the case $n = 2$ or $n = 3$. (If you've seen induction, try proving it for a general natural number n using Definition 3!) For any sets $A_1, A_2, \cdots, A_n$, we have that*

- $(A_1 \cup A_2 \cup \cdots \cup A_n)^C = A_1^C \cap A_2^C \cap \cdots \cap A_n^C$;
- $(A_1 \cap A_2 \cap \cdots \cap A_n)^C = A_1^C \cup A_2^C \cup \cdots \cup A_n^C$.

Note 6: Disjoint and Partition

Sets A and B are **disjoint** if they have no elements in common, that is, if $A \cap B = \emptyset$.
A **partition** of a set S is a collection of sets $A_1, A_2, \cdots, A_n \subseteq S$ such that $A_1, A_2, \cdots A_n$ are all pairwise disjoint (no two of the sets share an element) and $A_1 \cup A_2 \cup \cdots \cup A_n = S$.

Exercise 1.8. Let $S = \{1, 2, 3\}$. Write all possible partitions of S, and explain how you know that these are all the possibilities.

Exercise 1.9. Let S be the set of all cards in a standard 52-card deck. Let A be the set containing the clubs and the red face cards. Describe a set B such that A and B are disjoint. Do A and your set B form a partition of S? Explain.

Note 7: Bijection

A **bijection** between sets A and B is a function $f : A \to B$ such that for each element $b \in B$, there is exactly one element $a \in A$ such that $f(a) = b$.

Exercise 1.10. Describe or illustrate a bijective function $f : A \to B$ where $A = \{\text{red, blue, yellow}\}$ and $B = \{\text{shirt, skirt, headband}\}$.

Note 8: Cardinality

The **cardinality** of a set A, denoted $|A|$, is a measure of the size of the set. If A is the empty set, then $|A| = 0$. If there is a bijection between A and the set $\{1, 2, \cdots, n\}$ for some positive integer n, then $|A| = n$. (This means that for finite sets, the cardinality is the number of elements in the set.) In general, two sets have the same cardinality if there is a bijection between them.

Exercise 1.11. Suppose that A_1, A_2, and A_3 form a partition of a universal set S. Let B be some other set. If we know that $|B \cap A_1| = n$, $|B \cap A_2| = m$, and $|B \cap A_3| = k$, what is $|B|$ and why?

Note 9: Countable and Uncountable

A set A is called **countable** if it is either finite or in one-to-one correspondence with the natural numbers $\mathbb{N}$. (In other words, a set A is countably infinite if we can find some bijection between it and $\mathbb{N}$, matching each element of A to exactly one element of $\mathbb{N}$.)
A set is called **uncountable** if it is not countable. For example, the real numbers $\mathbb{R}$ are uncountable, as are all intervals such as $(0, 1)$.

Exercise 1.12. Find a bijection between the interval $(0, 1)$ and the interval $(3, 5)$.

Exercise 1.13. Using that $(0, 1)$ is uncountable, prove that any open interval (a, b) with $a < b$ and $a, b \in \mathbb{R}$ is uncountable. (Hint: generalize your work from Exercise 1.12.)

Theorem 1.3. *Let A and B be finite sets. Prove that $|A \cup B| = |A| + |B| - |A \cap B|$.*

Exercise 1.14. How would you generalize the previous theorem to give the cardinality of the union of three finite sets $|A \cup B \cup C|$ instead of just two? What about n finite sets?

Exercise 1.15. Suppose that a survey of Summer Olympics viewers in the US showed the following: 50% watched gymnastics, 80% watched track, 10% watched diving, 9% watched diving and gymnastics, 7% watched diving and track, 40% watched gymnastics and track, and 6% watched all three.

Find the percentage of the group that watched none of these three sports.

2 Laws of Probability

2.1 Introduction

Now that we know something about how sets work, we'll define probability and the axioms it follows. Most of this chapter will be working with those axioms and a few laws that result from them.

2.2 Definitions, Exercises, and Theorems

> **Note 10: Outcomes, Sample Space, and Events**
>
> Consider a procedure (called a **random experiment**) that has a well-defined set of possible results. We call each possible result of that experiment an **outcome**, and the set of all outcomes is the **sample space**, usually denoted S. An **event** is a subset of the sample space.

Exercise 2.1. What is the sample space if I roll a standard six-sided die? Describe in words an event A within this sample space that we might consider. How would you write that event as a subset of the sample space?

> **Tip 1: Axioms of Probability**
>
> Let S be a sample space. **Probability** is a function defined on events in that sample space, with the probability of event A denoted as $P(A)$. This function satisfies the following axioms:
>
> 1. For any event A, $P(A) \geq 0$.
> 2. $P(S) = 1$.
> 3. For mutually disjoint events $A_1, A_2, A_3, \cdots$, we have
>
> $$P\left(\bigcup_{i=0}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

Theorem 2.1. *Using the probability axioms, prove the following:*

- *For any event A, $P(A^C) = 1 - P(A)$.*
- *If $A \subseteq B$, then $P(A) \leq P(B)$.*

Exercise 2.2. Math, CS, Business, and English compete in a tournament, and exactly one of them will win. Business and CS have the same chance of winning the tournament. Math is three times as likely to win the tournament as English. The probability that either Math or Business wins the tournament is 0.7. Find the probabilities of each team winning the tournament.

Theorem 2.2. *Prove that in a finite sample space S with all outcomes equally likely, the probability of an event A is $P(A) = \dfrac{|A|}{|S|}$.*

Exercise 2.3. You roll a fair six-sided die twice. Let X_1 be the outcome of the first roll and X_2 the outcome of the second roll. Describe the sample space S as a set of ordered pairs (X_1, X_2). What outcomes are in the event A defined as the event that $X_1 + X_2 = 6$? What is the probability of A?

Exercise 2.4. The probability that a visit to a primary care physician's (PCP) office results in neither lab work nor referral to a specialist is 40%. Of those coming to a PCP's office, 45% are referred to specialists and 30% require lab work. Calculate the probability that a visit to a PCP's office results in both lab work and referral to a specialist[1].

Sample spaces can be countable, as in the examples above. We call these **discrete probability models**. In other cases, the sample space can be uncountable. We call these **continuous probability models**.

Exercise 2.5. Let T be the time in hours to complete a certain job at a factory. Based on historical data, we know that

$$P(T \leq t) = \begin{cases} \frac{1}{16}t^2 & \text{for } 0 \leq t \leq 4; \\ 1 & \text{for } t > 4. \end{cases}$$

Find the probability that the job needs **at most** 2 hours.

> **i** Note 11: Conditional Probability
>
> For two events A and B, the **conditional probability** of A given B, written $P(A|B)$, is the probability that event A occurs if we know that event B occurred. That is,

[1] From the Society of Actuaries' Exam P Sample Questions, question 2

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

Exercise 2.6. If I roll a fair six-sided die, what is the probability that I roll a 6? Now suppose that after I roll the die, I tell you that I rolled an even number. What is the updated probability that I rolled a 6?

(Note: In the language of conditional probability, this is the probability that I rolled a 6 given that I rolled an even number, or $P(6|\text{ even})$.)

Exercise 2.7. Suppose I flip two fair coins and don't show you the results. - What is the probability that both flips were tails?

- Suppose that I tell you that the first flip was a tail. Given that information, what is the probability that both flips were tails?
- Suppose that you ask me whether I got at least one tail, and I say yes. Given that information, what is the probability that both flips were tails?

i Note 12: Independence

We call two events A and B **independent** if $P(A \cap B) = P(A)P(B)$.

Exercise 2.8. Suppose I draw the top card from a shuffled, standard 52-card deck. Let A be the event that the card is a three and B be the event that the card is a club. Show that A and B are independent.

Exercise 2.9. Suppose I make a special card deck from a standard deck. I include only red 2, 3, and 4 cards and black 5 and 6 cards, and then I shuffle the deck and draw the top card. Let E be the event that the card is even and F be the event that it is red. Show that E and F are not independent.

Exercise 2.10. A washing machine is made up of many components. For simplicity, suppose that the ones that commonly break come in three categories: the motion system, timing control, and the water system. If any of those subsystems fails, the washing machine fails. Suppose the subsystems operate and fail independently of each other and that the probability of the motion system failing within ten years is 0.05, the probability of the control system failing within ten years is 0.015, and the probability of the water system failing within ten years is 0.07. What is the probability that the washing machine still works after ten years?[2]

[2]Adapted from Necip Doganaksoy, William Q Meeker, Gerald J Hahn, Product reliability: How statistics fits in, Significance, Volume 20, Issue 2, April 2023, Pages 28–32, https://doi.org/10.1093/jrssig/qmad029

Theorem 2.3. *Prove that if A and B are independent, then $P(A|B) = P(A)$.*

Exercise 2.11. Suppose you toss a pair of fair six-sided dice. Let A be the event that the sum of the rolls is a multiple of 2. Let B be the event that the sum of the rolls is a multiple of 3. Are A and B independent events? Why or why not?

Theorem 2.4 (Law of Total Probability)**.** *Prove that if $B_1, B_2, B_3, \cdots$ are a partition of the sample space S, then for an event A, $P(A) = \sum_i P(A|B_i)P(B_i)$.*

Exercise 2.12. Elise is playing a board game that involves rolling a fair six-sided die and spinning a spinner. If the die roll is a multiple of 3, the player spins a spinner with 4 equally-sized regions numbered 1 through 4. Otherwise, the player spins a spinner with 8 equally-sized regions numbered 1 through 8. The spinner result determines how far the player moves on the board. What is the probability that Elise will only move 1 spot on her turn?

Theorem 2.5 (Bayes' Rule)**.** *Prove that* $P(B|A) = \dfrac{P(A|B)P(B)}{P(A)}$.

Let's use Bayes' Rule a bit. The Law of Total Probability will also be helpful!

Exercise 2.13. About 50.5% of respondents to a survey about games were 45 years old or older. Of people ages 18-44, about 37.5% said they play board games at least monthly. Of people ages 45+, about 20.2% said they play board games at least monthly.[3]

A person who plays board games at least once a month is randomly chosen from the survey respondents. What is the probability that they are 45+ years old?

Exercise 2.14. A survey was conducted of those who say they contributed to charity during the past year. The survey results indicate that 80% of this population claims to have contributed at least \$500, 40% overstated the value of their contributions, and 38% did both. Calculate the probability that a randomly selected person overstated the value of their contribution, given that they claimed to have contributed less than \$500.[4]

Exercise 2.15. Small businesses in a particular city are categorized as retail, service, or other. In a study of yearly bankruptcies of small businesses in the city, this information was observed:

- 55% of the small businesses were retail, and of those, 10% went bankrupt.
- 30% of the small businesses were service, and of those, 12% went bankrupt.
- The rest of the businesses were other, and 5% of those went bankrupt.

An auditor randomly selected a small business that went bankrupt last year. Calculate the probability that it was a service business.[5]

[3] Based on a YouGov survey conducted in May 2026
[4] Adapted from Exam P Sample Questions, question 423
[5] Adapted from Exam P Sample Questions, question 370

Theorem 2.6. *Prove that if A and B are independent, then A^C and B are independent.*

Exercise 2.16. The Massachusetts Bay Transit Authority (MBTA) tracks and publishes data about the reliability of three lines (Blue, Orange, and Red) of Boston's subway system, the T. The MBTA tracks the percent of riders of that line who reached their destination within five minutes of the expected time. (We'll call those trips "on time.")

In one 12-month period, 16% of trips included in the data were on the Blue Line, 38% were on the Orange Line, and 46% were on the Red Line. Unfortunately, 9% of trips did not arrive on time. Of those, 14% were on the Blue Line, 21% were on the Orange Line, and the remaining 65% were on the Red Line.[6]

Marie-Caroline and Jocelyn each took separate trips included in the dataset. If Marie-Caroline's trip was on the Red Line, what is the probability that it did not arrive on time? If Jocelyn's trip was on time, what is the probability that it was on the Orange Line?

[6]Data from MBTA Performance Metrics, accessed on 6/5/2026

3 Counting

3.1 Introduction

In Theorem 2.2, you showed that for a finite sample space with equally likely outcomes, our probability can be found by looking at the number of outcomes in the event of interest. So we need a good way to count those outcomes. Counting might sound easy, but sometimes there's a lot of subtlety in what we're trying to count.

3.2 Definitions, Exercises, and Theorems

Exercise 3.1. As part of a game, I spin a spinner that is split equally among 3 colors (red, blue, and yellow), and I roll a fair four-sided die. How many combinations of spins and rolls could I get?

Exercise 3.2. In the scenario in Exercise 3.1, suppose the result I want is a red spin and a roll of either 3 or 4. What is the probability that I get my desired result?

Exercise 3.3. How many distinct sequences of heads and tails can I get from flipping a coin three times?

Exercise 3.4. Show that the number of four-digit PINs using digits 0 through 9 if digits can be reused is 10,000.

Exercise 3.5. How many four-digit PINs are there that do not use 2? What is the probability of choosing such a PIN from the set of all possible four-digit PINs?

Exercise 3.6. Suppose I have a set of n characters and need a password of length k. I am allowed to reuse characters. Find and explain an expression for the number of possible passwords.

Exercise 3.7. Suppose a license plate format consists of 3 numbers (with repeats allowed) followed by 3 letters (again with repeats allowed). How many license plates are possible?

Exercise 3.8. Now suppose that repeats are *not* allowed in the license plates described above. Each digit and each letter can be used at most once. How many license plates are possible?

Exercise 3.9. All eight students in a class are going to give individual presentations during finals period. In how many possible orders could they present?

> **Note 13: Factorial**
>
> The **factorial** is a useful shorthand: n-factorial, or $n!$, is defined as $n! = n \cdot (n-1) \cdots 2 \cdot 1$.

Exercise 3.10. On Thursday, only four of the students are going to give presentations, but they don't know in advance which four. How many possible ordered sets of four student presenters are there?

> **Note 14: Permutation**
>
> A **permutation** of n objects is a way of ordering n objects. A k-permutation from a set of size n is a way of ordering k objects chosen from a set of n objects. $P(n, k)$ denotes the number of k-permutations from a set of size n.

Theorem 3.1. *Explain why the number of k-permutations from a set of size n is $P(n, k) = \frac{n!}{(n-k)!}$.*

Exercise 3.11. The "bootstrap" is a statistical method for generating a new data set that is like an existing one. Suppose we have a data set consisting of 6 observations: $x_1, x_2, x_3, x_4, x_5, x_6$. To generate a bootstrap data set, we sample from the original data set with replacement, meaning that it is possible for each observation to be sampled more than once. An example bootstrap data set would be $x_2, x_3, x_4, x_2, x_1, x_3$. What is the probability that the bootstrap data set contains each observation exactly once?

Exercise 3.12. As I work in the evenings, I often listen to music. Suppose that I have a playlist of $n = 50$ songs and I listen to them on shuffle where the software always selects from the list with equal probability when selecting which song to play next. If I listen to $k = 10$ songs, what is the probability that at least one of the songs will be duplicated? What about if I listen to $k = 30$ songs? (Hint: think about the complement of the desired probability!)

Exercise 3.13. There are p people in the room. Assume that each person's birthday is equally likely to be any of 365 days (ignore leap years!) and that everyone's birthdays are independent. What is the probability that no one in the room has the same birthday? What is the least value of p for which there is at least a 50% chance that two people do share a birthday?

Exercise 3.14. Let $S = \{1, 2, 3, 4, 5\}$. Answer each of the following:

a. How many ordered pairs with elements from S are there?

b. How many ordered triples with elements from S are there?

c. How many different two-element subsets of S are there? (Recall that order does not matter in sets.)

d. How many different three-element subsets of S are there?

e. How are the answers to (a) and (c) related? What about (b) and (d)?

Theorem 3.2. *Explain why the number of ways to pick a subset of size k from a set of size n is* $\dfrac{n!}{k!(n-k)!}$.

> **Note 15: Combination**
>
> The **combination** $C(n, k)$ or $\binom{n}{k}$, usually read "n choose k," is the number of ways to pick k unordered things from a set of size n.

Exercise 3.15. As part of a card game, I am dealt three cards from a standard 52-card deck. How many possible hands could I have? What is the probability that my hand does not contain an ace? (There are four aces in the deck.)

Exercise 3.16. How many distinct sequences could I make using two copies of the letter H and two copies of the letter T?

Exercise 3.17. I flip an unfair coin with $P(H) = p$ four times. Find the probability that I flip exactly two tails. (Hint: What is the probability of the outcome $TTHH$? What about $HTHT$?)

> **Note 16: Bernoulli Trials and Binomial Experiments**
>
> A **Bernoulli trial** is a random experiment with two possible outcomes: success with probability p or failure with probability $1 - p$. A **binomial experiment** is a series of n independent Bernoulli trials in which we count the number of successes.

Theorem 3.3. *Explain why the probability of k successes in a binomial experiment is*

$$P(k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Exercise 3.18. A woman, Joy, claimed that she could smell Parkinson's disease. Researchers gave her 12 shirts and asked her to identify which ones had been worn by someone with Parkinson's. Initially, researchers believe that she correctly identified 11 of the shirts. If she had been randomly guessing, what is the probability that she would have identified **at least** 11 of the shirts correctly? Based on this, do you believe her claim or not? (When you're done with the problem, I'll tell you what happened next!)

Exercise 3.19. A particular tennis player has a first serve percentage (which means his attempt at serving the ball lands in a valid place) across his career of 65%. In one set, 17 of his 18 first serves were valid. What is the probability of this player serving at least this well across 18 first serves?

Exercise 3.20. A club needs to choose two co-presidents. The names of all the current members are put into a hat, and the first two that are drawn will be co-presidents. The names are not replaced once they are drawn (one person cannot be two presidents). You want to know the probability that the presidents both have the same major. Is this binomial, and why/why not?

3.3 Challenge Questions

Exercise 3.21. The game SET involves a deck of 81 cards. Each card has a picture of some number (one, two, or three) of shapes (oval, squiggle, or diamond), which is painted in one color (green, red, or purple), with some shading (solid, hatched, or empty). Each number-shape-color-shading combination appears in the deck exactly once. Three cards are called a set if for each attribute, either all three cards match or all three are different. (For example, for color, either all three cards need to be the same color, or they each need to be a different color.) How many different sets are in the deck? What is the probability of there being a set in the first 4 cards dealt from a well-shuffled deck?

Exercise 3.22. A mail carrier delivers mail to nineteen houses on the east side of Blossom Street. The carrier notes that no two adjacent houses ever get mail on the same day, but that there are never more than two houses in a row that get no mail on the same day. How many different patterns of mail delivery are possible?[1]

[1] From AIME 2011

4 Introduction to Discrete Random Variables

4.1 Introduction

So far, we've looked at how the probability function works in general and have thought about counting equally likely outcomes. Here, we'll add some mathematical language and machinery for working with outcomes and their probabilities. We'll also look at some other ideas related to probability that we might be interested in, like measures of a typical outcome and how much outcomes vary.

4.2 Definitions, Exercises, and Theorems

> **Note 17:** Random Variable
>
> A **random variable** X is a function from the sample space to the real numbers. That is, $X : S \to \mathbb{R}$.

Here are some examples of random variables: the sum of rolls of two dice, the number of heads in a sequence of 8 coin flips, the total winnings from playing a particular probability-based game, the number of hats given to the correct people when n people are handed their hats back at random.

Exercise 4.1. Find the possible values (ranges of the functions) for each of these random variables:

a. The number of heads in three coin flips

b. The time until you next receive an email

c. The number of flips of a coin until either the first heads appears or n flips occur

d. The number of flips of a coin until the first heads appears

e. The time until a cell phone is either plugged into a charger or dies

f. The number of numbers 7 or above when four numbers from 0-9 are selected randomly, with repeats allowed

g. The number of numbers 7 or above when four numbers from 0-9 are selected randomly, with repeats not allowed

h. The time until the next occurrence of rain in a particular city

Exercise 4.2. Write the function definition for the random variable in Exercise 4.1 (a). In other words, find the sample space and map each outcome to a real number based on this random variable.

We'll separate random variables into multiple types, and we'll talk about two of those types: discrete and continuous.

> **Note 18: Discrete Random Variable**
>
> A **discrete random variable** is a random variable with a countable range. That is, it can only take on countably many values.

We'll formally define a **continuous** random variable later, but for the moment, consider them to be random variables with uncountably infinite ranges.

Exercise 4.3. Which random variables from Exercise 4.1 are discrete?

Okay, so we have discrete random variables. How do we start talking about probabilities associated with them?

> **Note 19: Probability Mass Function**
>
> Let X be a discrete random variable with range $\{x_1, x_2, x_3, \cdots\}$ The **probability mass function** (or PMF) P_X assigns a probability to each value in the range. That is, $P_X(x_k) = P(X = x_k)$ for every x_k in X's range.

Exercise 4.4. Find the probability mass function for the random variable you defined in Exercise 4.1 (a) and Exercise 4.2.

Exercise 4.5. I have a fair six-sided die which I roll until I roll a 6 for the first time. Let X be the total number of die rolls. Find the probability mass function of X.

The PMF isn't the only useful way to talk about probabilities associated with a discrete random variable.

> **Note 20: Cumulative Distribution Function**
>
> The **cumulative distribution function** (or CDF) of a random variable X is $F_X(x) = P(X \leq x)$ for all $x \in \mathbb{R}$.

Exercise 4.6. Find the cumulative distribution functions for the random variables you studied in Exercise 4.4 and Exercise 4.5.

(Hint: For the random variable from Exercise 4.5 the expression for the sum of a geometric sequence might be helpful. Feel free to look this up! You could alternatively think about the complement of what you're looking for.)

Exercise 4.7. Find expressions in terms of F_X and P_X for $P(a < X \leq b)$ and $P(X < x)$ if X is a discrete random variable.

It's often useful to think about the "typical" or **expected** value of a random variable. Let's define that formally.

> **Note 21: Expected Value**
>
> The **expectation** or **expected value** of a random variable X with probability mass function P_X is $E[X] = \sum_k x_k P_X(x_k)$, that is, a weighted average of the values in the range of X.

Exercise 4.8. Find the expected value of a roll of a fair n-sided die.

Exercise 4.9. A professor bakes cookies for finals week. The probability that she bakes 3 batches of cookies is 0.25, the probability that she bakes 4 batches of cookies is 0.3, the probability that she bakes 5 batches of cookies is 0.35, and the probability that she bakes 6 batches of cookies is 0.1. Find the expected value of the number of batches of cookies she bakes.

Exercise 4.10. A math camp of 120 students is driven in 3 buses to an amusement park. There are 36 students in one of the buses, 36 in the second bus, and 48 in the third bus. (Many students run late…) When the buses arrive, one of the 120 students is randomly chosen. Let X denote the number of students on the bus of that randomly chosen student, and find $E[X]$. Is this less than, equal to, or greater than the average number of students per bus, and why?

Exercise 4.11. The class size paradox states that students tend to have a larger average class size than a school's stated average class size. Explain this using your result from the last exercise.

Exercise 4.12. In the carnival game chuck-a-luck, three dice are rolled. You make a bet on a particular number (1, 2, 3, 4, 5, 6) showing up. The payout is 1 to 1 (that is, \$1 of net winnings for every \$1 you bet) if that number shows on (exactly) one die, 2 to 1 if it shows on two dice, and 3 to 1 if it shows up on all three. You lose your initial stake if your number does not show on any of the dice. If you make a \$1 bet on the number three, what is your expected net winnings?[1]

Some random variables are functions of other random variables. The next result is one of two results that tell us that expectation is **linear**, or in other words, expectation behaves nicely with linear combinations of random variables. (We'll need more tools to get to the second result.)

Theorem 4.1. *Prove that for any $a, b \in \mathbb{R}$, $E[aX + b] = aE[X] + b$.*

Exercise 4.13. Consider a random variable X with $P_X(x_k) = \frac{1}{7}$ for $x_k \in \{-3, -2, -1, 0, 1, 2, 3\}$. Let another random variable $Y = X^2$. Find the range of Y and the probability mass function P_Y. Using P_Y, find $E[Y]$.

Theorem 4.2 (Law of the Unconscious Statistician (LOTUS))**.** *Let X be a random variable, and let $Y = g(X)$ for some function g. Then we have $E[Y] = \sum_k g(x_k)P_X(x_k)$. While this holds in general, prove it for the case where the range of X is finite.*

Now we'll think about measuring the spread or variation of the possible values of X. The measures we'll use are related to the distance between values of X and its expectation.

> **i** Note 22: Variance and Standard Deviation
>
> Let X be a random variable with expectation $E[X]$. Then the **variance** of X, denoted $\text{Var}(X)$, is given by $\text{Var}(X) = E[(X - E[X])^2]$. The square root of variance is called **standard deviation**, or $\text{SD}(X)$.

Exercise 4.14. Find $\text{Var}(X)$ if X is the outcome when a fair six-sided die is rolled.

The definition of variance isn't the easiest to actually do calculations with. Luckily, there's an easier way. Let's see it and prove that it works:

Theorem 4.3. *Prove that* $\text{Var}(X)$ *can be written as* $E[X^2] - (E[X])^2$.

Exercise 4.15. Recalculate the variance in Exercise 4.14 using the form of variance from Theorem 4.3.

[1] Problem from Dennis Sun.

Theorem 4.4. *Prove that* $\text{Var}(aX + b) = a^2\text{Var}(X)$.

Exercise 4.16. In terms of $\text{SD}(X)$, what is $\text{SD}(aX + b)$ where a, b are real numbers with $a \neq 0$?

Exercise 4.17. A box contains five red and five blue marbles. Two marbles are withdrawn randomly. If they are the same color, you win \$1.10, and if they are different colors, you lose \$1.00. Find the expected value of the amount you win and the variance of the amount you win. (Please do not use this to start a casino on the quad.)[2]

Exercise 4.18. Find $\text{Var}(X)$ and $\text{Var}(Y)$ from Exercise 4.13.

Exercise 4.19. Let X be a discrete random variable with range $\{0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi\}$ such that $P_X(x_i) = \frac{1}{5}$ for all x_i in the range. Find $E[\cos(X)]$ and $\text{Var}(\cos(X))$.[3]

Exercise 4.20. Suppose the annual cost of maintaining and repairing a car in Fitchburg has an expected value of \$500 and a variance of 280 square dollars[4]. A tax of 25% is introduced on all items associated with the maintenance and repair of cars, so everything is now 25% more expensive. Now what are the expected value and variance of annual car maintenance and repair?[5]

4.3 Further Practice

Exercise 4.21. Toma Todo is a game played by spinning a top called la pirinola and then following its instructions. The game starts with each player having a certain number of tokens (beans, chips, etc.) and a certain number of tokens in the central pot. The pirinola has six sides: Toma Todo (take all the chips from the pot), Toma 2 (take 2 chips from the pot), Toma 1 (take 1 chip from the pot), Pon 1 (put 1 chip in the pot), Pon 2 (put 2 chips in the pot), and Todos Ponen (all players put 2 chips in the pot).

On a particular turn, what is the expected value of the number of chips a player will win? What is the variance?[6] (*Note: you will need to introduce an assumption or some notation.*)

[2] Problem from probabilitycourse.com. A former student really, really wanted to use this game to start a terrible casino on the quad.
[3] Problem from probabilitycourse.com
[4] The main problem with variance is the units. What even is a square dollar?
[5] Modified from a Sample Exam P problem.
[6] Problem from Larry Lesser's presentation at eCOTS 2022.

4.4 Challenge Questions

Exercise 4.22. Let X be a random variable, and let $Y = e^X$. (In function form, this means that $Y(x) = e^{X(x)}$.) Find the cumulative distribution function F_Y in terms of F_X. (In other words, if $F_Y(y) = F_X(x)$, how are y and x related?)

Exercise 4.23. Consider the following game. You roll a fair die once and observe the result X. At this point, you can stop the game and win X dollars. You can also choose to roll the die for the second time to observe the value Y. In this case, you will win Y dollars. Let W be the value that you win in this game. What strategy do you use to maximize $E[W]$? What is the maximum $E[W]$ you can achieve using your strategy? What is $\text{Var}(W)$ using your strategy?[7]

Exercise 4.24. You and a friend are given two identical envelopes. You know that one envelope contains twice as much money as the other. You are offered the opportunity to switch envelopes with your friend. What is the expected value of the money in the other envelope in terms of the money in your envelope? Based on this, should you switch? Something is a little strange about this scenario. Can you resolve the strangeness?

[7] Modified from probabilitycourse.com.

5 Common Discrete Random Variables

5.1 Introduction

Now that we've looked at general properties and uses of discrete random variables, we'll investigate some types of discrete random variables that are often useful.

5.2 Definitions, Exercises, and Theorems

We'll start with a result that we **will not prove** at this time. The first part is the other aspect of what it means for expectation to be linear. The second part is a useful fact about variance.

Theorem 5.1. *Given two random variables X_1 and X_2, $E[X_1 + X_2] = E[X_1] + E[X_2]$.*

If random variables X_1 and X_2 are independent, then $\text{Var}(X_1 + X_2) = \text{Var}(X_1) + \text{Var}(X_2)$.

We haven't formally defined independence for random variables yet. For now, we can use this to represent independent trials of the same random process.

Now we'll move into the types of random variables. The first two will be familiar from our discussion of counting.

> **Note 23:** Bernoulli Random Variable
>
> A **Bernoulli random variable** with parameter p, written as $X \sim \text{Bernoulli}(p)$, has successes (with $X = 1$) and failures (With $X = 0$) and a probability mass function given by
>
> $$P_X(k) = \begin{cases} 1-p & k = 0 \\ p & k = 1 \end{cases}.$$

> **Note 24:** Binomial Random Variable
>
> A **binomial random variable** with parameters n and p, written as $X \sim \text{Binomial}(n, p)$, is n independent Bernoulli trials, with $P_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$.

You already proved this expression for the PMF of a binomial random variable in Theorem 3.3.

Exercise 5.1. Find the expectation and variance of a Bernoulli random variable in terms of p.

Exercise 5.2. A basketball player who makes 90% of free throw attempts has 8 attempts in a game. What is the probability of this player making exactly 7 of those free throws? 7 or more?

Exercise 5.3. A communication system consists of n components, each of which will independently function with probability p. The total system will be able to operate if at least half of its components function. For what values of p is a 5-component system more likely to operate than a 3-component system? (*Hint: write the inequality that would let you find this, and then feel free to use a computer to solve it.*)

The next lemma will help us find the expectation of a binomial random variable.

Theorem 5.2. *Prove that the expectation of a binomial random variable is np and that the variance is $np(1-p)$. (Hint: use Theorem 5.1.)*

Binomial random variables aren't the only family that we can build based on Bernoulli trials.

> **i** Note 25: Geometric Random Variable
>
> A **geometric random variable** with parameter p, or $X \sim \text{Geometric}(p)$, is the total number of independent Bernoulli trials to achieve the first success. It has probability mass function $P_X(k) = p(1-p)^{k-1}$.

Exercise 5.4. Consider the basketball player from Exercise 5.2. What is the probability that it would take that player exactly 2 attempts to make their first free throw of the game? exactly 3 attempts?

Exercise 5.5. Use the sum of a geometric series to show that the geometric random variable's PMF does indeed sum to 1 over all possible values of k.

Theorem 5.3. *For a geometric random variable X, prove that $E[X] = \frac{1}{p}$. (Hint: One way to approach this is to take a derivative with respect to r of a geometric series. Another way is to split the sum into many similar sums that start at different values.)*

Theorem 5.4. *For a geometric random variable X, prove that $\text{Var}(x) = \frac{1-p}{p^2}$. (Hint: one approach is to use that $E[X^2] = E[X(X-1)] + E[X]$ and use the second derivative of a geometric series.)*

Exercise 5.6. Go back once more to the basketball player from Exercise 5.2. How many attempts X would we expect them to need to make their first free throw of the game? What is the variance of X?

We're not always interested in just the first success. We can expand this idea a bit more.

> **Note 26: Negative Binomial Random Variable**
>
> A **negative binomial random variable** or **Pascal random variable**, $X \sim \text{NB}(m, p)$, is the number of independent Bernoulli trials with probability of success p until there have been m successes.

Theorem 5.5. *Explain why, for* $X \sim \text{NB}(m, p)$, $P_X(k) = \binom{k-1}{m-1} p^m (1-p)^{k-m}$.

Exercise 5.7. What's the probability that we need to roll a fair six-sided die 10 times to get three rolls of 2?

Prove only the expected value portion of the following:

Theorem 5.6. *The expected value of a negative binomial random variable* $\text{NB}(m, p)$ *is* $\frac{m}{p}$, *and the variance is* $\frac{m(1-p)}{p^2}$. *(These should seem related to the expectation and variance of a geometric variable!)*

Exercise 5.8. A company needs to hire four robot operators, and they will hire the first four applicants that are suitable. The probability that any given applicant is suitable is 0.8. Find the probability that the company interviews more than 6 applicants before filling all the jobs.

We can also use the success/failure structure a different way:

> **Note 27: Hypergeometric Random Variable**
>
> A **hypergeometric random variable**, $X \sim \text{Hypergeometric}(b, r, k)$, is the number of successes in an experiment where k items are chosen without replacement from a group of $b + r$ things, b of them successes and r of them failures.

Theorem 5.7. *Explain why, for* X *a hypergeometric random variable,* $P_X(z) = \dfrac{\binom{b}{z}\binom{r}{k-z}}{\binom{b+r}{k}}$.

Exercise 5.9. A process known as capture-recapture is often used to estimate animal populations or density. Suppose that 20 ocelots live in a particular wetland area. You initially capture, mark, and release 5 of them. A few months later, you capture and then release 6 ocelots. What is the probability that none of the ocelots were already marked?

We can also think about how many times something occurs in a given amount of time or space as opposed to a certain number of discrete trials.

Note 28: Poisson Random Variable

A Poisson (pronounced pwah-sawn) random variable $X \sim \text{Poisson}(\lambda)$ tells us how many times a certain event occurs in a given interval of time/space if λ is how often it occurs on average. The probability mass function is $P_X(k) = \dfrac{e^{-\lambda}\lambda^k}{k!}$.

Exercise 5.10. Suppose that the number of emails I receive on weekends can be modeled as a Poisson distribution with $\lambda = \frac{1}{30}$ emails per minute. What is the probability that I receive no emails between 8am and noon on Sunday? At most two emails?

Exercise 5.11. Suppose that the number of typos in one of these chapters has a Poisson distribution with parameter $\lambda = \frac{1}{2}$. Calculate the probability that there is at least one error in this chapter.

Theorem 5.8. *Prove that for $X \sim \text{Poisson}(\lambda)$, the expectation $E[X] = \lambda$ and the variance $\text{Var}[X] = \lambda$. (Note: look out for the Taylor series expansion of e^λ.)*

6 Summary

In summary, this book has no content whatsoever.

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1 + 1
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[1] 2
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References